Big Data Infrastructure Using Social Media Insights and Developing a Global Wine Trading Platform

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ABSTRACT

This paper examines the role of big data platforms in various real-world applications, presented in two distinct parts. The first part addresses the extraction and processing of live posts from the Reddit API, using Spark Streaming to manage real-time data flows. This data is deployed on AWS through Kinesis, enabling real-time sentiment analysis which is then visualized with tools like AWS QuickSight.

The second part focuses on the creation of a global wine trading platform, designed to streamline the transport of wine from Cape Town to California. By integrating flight, traffic, and weather APIs, this platform optimizes logistics to ensure timely delivery, with wine arriving at restaurants one hour before opening. Together, these parts demonstrate the impact of big data and API-driven platforms on everyday operations.

KEYWORDS

kinesis, aws, reddit, api, spark, wine, big data, ec2, pyspark, athena, pyspark

1 INTRODUCTION

In today's data-driven world, effective big data infrastructure and platforms are essential for processing, analysing, and deriving actionable insights from vast data sources. This report presents two distinct but complementary projects that demonstrate the potential of big data solutions in diverse applications.

Group Project 1 focuses on building a scalable big data infrastructure capable of handling both real-time (hot path) and batch (cold path) data. Leveraging the lambda architecture. The project integrates data sources reddit application programming interface (API) and APIs for airlines, weather and traffic. Advanced tools, including PySpark and Apache Spark, enable sophisticated data mining and machine learning tasks. This infrastructure supports real-time and batch processing, accommodating use cases like airline monitoring, weather prediction, and traffic congestion analysis to illustrate the versatility and impact of robust data management systems.

Group project 2 involves designing a global wine trading platform to optimize wine delivery logistics and streamline the supply chain. The platform connects wine producers with restaurants worldwide, incorporating real-time data sources, smart trading mechanisms, and logistics optimization to ensure timely delivery. By integrating API data for flights, traffic, and weather, and leveraging big data tools like Hadoop and Apache Spark, the platform provides real-time and batch processing capabilities essential for auction matching, logistics planning, and financial analysis. Together, these projects exemplify the power of big data and machine learning to tackle real-world challenges across various industries, from social media analytics to global trade logistics.

In conclusion, the rapid evolution of big data and IoT technologies has transformed various industries, offering new capabilities for real-time analytics, predictive insights, and efficient decision-making.

2 LITERATURE REVIEW

The exponential growth of data has made it critical to design robust architectures capable of processing massive amounts of information in real time. Big data architectures, such as lambda and kappa, have been instrumental in handling both real-time (hot path) and batch (cold path) data processing. Marz and Warren [1] highlight that lambda architecture is particularly effective for applications that require high-speed data processing alongside accurate, historical data analytics. Kappa architecture, as introduced by Kreps [2], offers a streamlined alternative by focusing solely on real-time data, which reduces complexity and maintenance requirements. These architectures form the backbone of group project 1, enabling simultaneous analysis of live and historical data for applications that demand immediate insights.

The integration of IoT (Internet of Things) devices has further expanded the scope of big data by enabling the continuous capture of data from diverse sources such as sensors, vehicles, and connected appliances. Wang et al. [3] emphasize that IoT devices create dynamic networks capable of supporting various applications, from environmental monitoring to predictive maintenance. In the context of Group Project 1, IoT sensors provide essential real-time data streams that enhance the

infrastructure's capability to analyse immediate environmental factors, including pollution levels and traffic patterns. By incorporating IoT data, big data architectures can support more comprehensive and responsive analytics, accommodating the needs of a modern, interconnected world.

With the rising demands for real-time insights, machine learning (ML) has emerged as an essential tool in big data ecosystems.

Gandomi and Haider [4] explain that integrating ML with frameworks like Apache Spark enables data scientists to apply advanced predictive analytics on large datasets effectively. Apache Spark's MLlib, a machine learning library specifically designed for scalable data processing, provides critical functionalities for applications in classification, clustering, and sentiment analysis. These capabilities are key to project 1's infrastructure, which utilizes ML to analyze social media posts and IoT data streams, generating timely insights that inform decision-making across sectors.

Social media platforms, especially Twitter and Reddit, have become indispensable sources of public sentiment and opinion, reflecting trends and social responses in real time. As Giachanou and Crestani [5] illustrate, sentiment analysis on social media offers a unique lens through which public perception of events or brands can be measured instantly. Group Project 1 leverages the Reddit API for live data streaming, which allows for real-time sentiment analysis through Spark Streaming on AWS infrastructure. This analysis supports various applications, from marketing to emergency response, by providing immediate feedback on public sentiment.

In the realm of logistics and supply chain management, big data has transformed traditional methods by enhancing efficiency, forecasting, and route optimization. Christopher [6] notes that logistics management now benefits from real-time data on traffic, weather, and transportation delays, which enable companies to optimize delivery schedules and routes. This data-driven approach is essential to Group Project 2, which aims to build a wine trading platform that coordinates deliveries from Cape Town to California by monitoring conditions and predicting the fastest, safest routes. Optimizing logistics through data insights not only improves operational efficiency but also aligns with evolving customer expectations for timely deliveries.

API integration is a foundational element in data-driven applications, offering structured access to diverse real-time data sources such as traffic, weather, and transportation information. Fielding [7] highlights that APIs support a modular and consistent approach to data access, facilitating the development of applications that require external data streams. In Project 2, APIs for flights, road traffic, and weather provide real-time insights that enable accurate tracking and routing of wine shipments to ensure on-time delivery. This seamless API integration is essential for applications that require continuously updated information to respond proactively to changing conditions.

As global supply chains become increasingly complex, big data tools offer new solutions to manage logistics across borders, predict demand, and assess risk. Xu et al. [8] assert that big data enhances global supply chain management by allowing organizations to identify and mitigate risks in real time. In the context of group project 2, data analytics from APIs and other sources assist in forecasting demand, identifying the best shipping routes, and managing cross-border delivery schedules. These capabilities are crucial in ensuring that wine shipments meet the project's strict one-hour delivery window, thereby illustrating the role of big data in optimizing international supply chains.

Finally, data visualization tools are indispensable for interpreting and communicating insights derived from big data. Murray [9] discusses the value of visualization tools like AWS QuickSight and Tableau, which allow users to explore data interactively and in real time. For both group projects, visualization is crucial to monitor and interpret complex data streams, such as public sentiment trends from social media or logistics data for wine shipments. These visual tools help stakeholders make informed, data-driven decisions swiftly, showcasing the importance of accessible data insights in real-time application.

3 PROJECT PROPOSAL

For the first project, the approach started with gathering Reddit posts in real time by utilizing the Reddit API. To access the data, it was necessary to sign up for the Reddit developer portal, where API keys were generated for programmatic access. This allowed the collection of real-time posts that would be used for sentiment analysis and other forms of processing. By extracting the posts via the API, we could ensure a continuous stream of fresh data for further analysis, enabling the system to react to trends and topics as they evolved.

Once the data was collected from Reddit, it was streamed in real time using AWS Kinesis. Kinesis provided the necessary infrastructure to handle the large volume of incoming data, ensuring that the posts were processed without delays. Kinesis allowed the data to be ingested and transmitted efficiently, which was crucial for ensuring the scalability and performance of the system. This step was integral in maintaining a smooth flow of data from the source to the processing layers in the pipeline.

After the data was ingested via Kinesis, it was stored in a NoSQL DynamoDB database, which was ideal for handling the large and varied data types from Reddit posts. DynamoDB's flexibility in managing semi-structured data allowed for easy storage and retrieval, while also providing the high performance needed for real-time operations. The seamless integration between DynamoDB and other AWS services ensured that the data could be accessed and processed in an optimized manner. The NoSQL structure facilitated quick access and allowed for scaling as the volume of posts increased.

To enable efficient querying of the data, Amazon Athena was configured to work with DynamoDB. Athena served as the query engine for DynamoDB, allowing users to run SQL-like queries on the stored data without needing to set up complex infrastructure. This integration ensured that users could perform quick and easy queries on large datasets, making the process of accessing insights from the

Reddit posts much more efficient. Athena's ability to work seamlessly with DynamoDB helped automate the querying and reporting process.

Finally, the results from the processed data were stored in an S3 bucket, from which they would be used to feed machine learning models in AWS. These models would be used to analyse the data and extract useful patterns, such as sentiment trends or topic modelling. Additionally, the data stored in the S3 bucket would be used to build dynamic dashboards in Amazon QuickSight. QuickSight would provide visualization tools to monitor the real-time data and insights generated by the system, making it easier for stakeholders to interpret and act on the data efficiently.

4 PROJECT IMPLEMENTATION

This proposal outlines two data-driven projects: the first establishes a big data infrastructure to capture and analyse Reddit posts in real time using AWS tools like Kinesis, DynamoDB, and Athena, feeding insights into ML models and visual dashboards. The second project focuses on building a global wine trading platform, using real-time data from APIs (flights, traffic, weather) to optimize logistics and ensure timely deliveries. Both projects leverage scalable AWS architecture, with project 1 emphasizing social media analytics and project 2 enhancing international supply chain efficiency. Together, these projects demonstrate the power of big data in diverse applications.

4.1 Real-Time Social Media Data Pipeline for Insights Using AWS

The platform for project 1 was designed to ingest Reddit data in both streaming and batch formats, ensuring flexibility and responsiveness in data handling. Reddit was integrated as the primary data source using the praw library, allowing the platform to retrieve posts from specific subreddits alongside metadata like titles, scores, URLs, and comment counts. To facilitate real-time data processing, Amazon Kinesis, managed via the boto3 library, handled the high-throughput ingestion of Reddit posts, streaming them for near-instant analysis and insight generation. Batch data, on the other hand, was stored in AWS S3, providing a reliable storage solution for long-term analysis and historical comparisons.

To support batch processing and unified analysis, multiple CSV files containing Reddit posts from various subreddits were loaded into separate data frames using pd.read_csv(). These files covered a wide array of topics, including Anxiety, Business, College, and Science, providing a broad data set for exploration. The individual dataframes were then merged into a single dataframe, merged_df, using pd.concat(). This consolidated dataframe allowed for efficient data processing and analysis across all subreddits, enabling cross-topic trend analysis.

Data cleaning steps were applied to merged_df to ensure consistency and quality. Specifically, the is_original column was renamed to avoid duplication issues, and unnecessary columns, such as Unnamed: 0 and is_original_2, were dropped. After cleaning, the data was converted into a Spark Dataframe, reddit_posts, for distributed storage and analysis, leveraging Spark's scalability for processing large datasets. This Spark Dataframe was saved as a table in the Hadoop Distributed File System (HDFS) under dev_ds_playground.social_posts, supporting future exploratory data analysis (EDA).

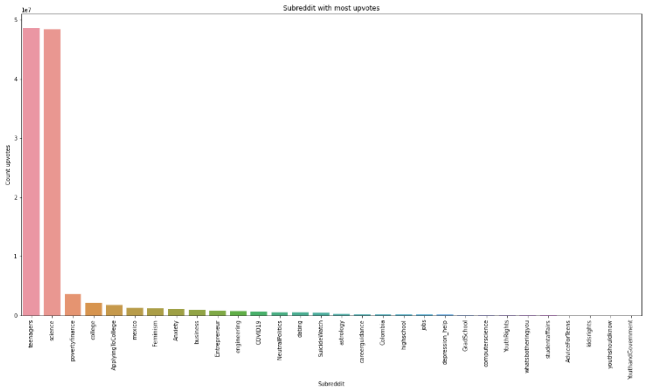

Figure 1: Subreddit with most upvotes

To continue processing, the data was then loaded into a Spark data frame directly from the pre-processed CSV in S3, facilitating efficient handling of large datasets in a distributed manner. Spark was initialized, and the data from S3 was ingested to inspect its schema and sample content, enabling the platform to quickly assess the data structure and quality. This setup provided a foundation for the next steps of model training and deployment.

For EDA, the cleaned and structured dataset was used to generate visualizations and uncover patterns within the data. These visuals provided insights into trends across different subreddits, helping to identify topics with high engagement, sentiment patterns, and other critical insights that could inform further analysis or model development. The platform's design allows for ongoing analysis, with the potential for real-time updates as new data is streamed in via Amazon Kinesis, providing an infrastructure for continuous Reddit data exploration.

To further enhance data accessibility and visualization, I established an architecture for integrating DynamoDB, Athena, and Amazon QuickSight, which followed these steps. We first created a data source in Amazon Athena named *Dynamo_reddit*, specifically configured to connect to DynamoDB, enabling seamless data access for querying. In Athena's Connection Details, we used a pre-existing template to create an AWS Lambda function that serves as an ETL tool, extracting data from DynamoDB and creating a corresponding table in Athena.

Once the Lambda function was created and executed, the DynamoDB table became accessible in Athena, allowing SQL queries for deeper data insights. To ensure data was stored for easy analysis, we developed an additional Lambda function that runs queries on the Athena table and stores results in an Amazon S3 bucket. This S3 storage, updated in real-time, was maintained

through a CloudWatch rule, triggering the Lambda function each time new data was added to DynamoDB.

Finally, we used Amazon QuickSight to visualize the S3-stored data, enabling insights from real-time information to support data-driven decisions. A logistic regression model was then chosen to classify user sentiment based on features like upvotes, comment length, and reply count. Spark's VectorAssembler was used to prepare these features for a logistic regression model, which was trained on labelled sentiment data to predict positive or negative sentiment.

Figure 2: Hot path insights in QuickSight

Predictions and associated probabilities were generated for each post, allowing for comprehensive model evaluation based on accuracy, precision, and recall. The initial model performed well, with balanced precision and recall and an anticipated accuracy of 70-80%. The trained model was saved back to S3, making it accessible for future deployment in real-time or batch sentiment analysis.

Future platform improvements could involve further model optimization, expanding feature sets with additional text-based attributes, and implementing real-time prediction services for ongoing sentiment tracking and monitoring. This approach equips the platform to deliver comprehensive insights into Reddit user sentiment and anomaly trends across diverse topics.

To ensure continuous integration and deployment, I implemented a CI/CD pipeline combining GitHub Actions and AWS CodePipeline. This setup automates the deployment and management of the data pipeline, handling both batch and real-time processing. Amazon Kinesis is employed for both streaming data and time-limited data handling, ensuring that real-time data is consistently updated. GitHub Actions automates the deployment by triggering workflows that identify updated scripts and push changes to the main branch. AWS CodePipeline, working alongside GitHub Actions, continuously monitors code updates, testing and deploying scripts directly to AWS services like Kinesis, Lambda, DynamoDB, and S3 for a seamless data flow.

Real-time processing begins with data ingestion from the Reddit API, which is streamed into Amazon Kinesis. As data arrives, AWS Lambda functions process it in real time, performing tasks such as cleaning and initial data modifications before storing the data in DynamoDB. Batch processing is handled by Apache Spark, which processes data in DynamoDB and saves summarized results to S3, enabling long-term storage and further analysis using Amazon Athena.

In addition to automating deployments, this CI/CD setup includes alerts, error reporting, and continuous monitoring to maintain a stable and manageable pipeline. This combination of tools and processes provides a robust framework for handling high-volume Reddit data and supports both real-time insights and in-depth historical analysis.

4.2 Integrated Data Collection System for Wine Demand and Supply Using AWS

The project aimed to create a data collection system for gathering and organizing information from restaurant websites and airline APIs to inform insights into customer preferences and travel behaviour. This system was designed with two core data sources: first, restaurant data, which included locations, operating hours, wine varieties, and tasting scores, and second, real-time airline data covering flight schedules, pricing, and routes to allow for data analysis that aligns with travel trends impacting restaurant operations.

To meet the project's data collection requirements, specific technologies for web scraping and API integration were utilized. Python libraries, including BeautifulSoup, Scrapy, and Selenium, were employed for their capabilities in parsing HTML and handling dynamic content, essential for retrieving structured information from restaurant websites. Meanwhile, the Amadeus API was integrated for accessing real-time flight data, supporting the continuous update of travel data critical to understanding demand fluctuations in restaurant operations.

A major component of the project involved a structured web scraping process focused on extracting key data fields from restaurant sites, such as wine names, brands, tasting scores, and operating hours. This process allowed for an organized compilation of information on consumer wine preferences and potential restaurant demand patterns. Airline data was simultaneously collected through scheduled API requests, which stored relevant flight information in CSV format for streamlined access and analysis.

To effectively categorize wine data based on consumer demand, a rule-based approach was developed. Wines were sorted according to price, type, and tasting score into distinct demand levels, including high, moderate, low, budget-friendly high demand, and standard demand. For example, wines with a tasting score above 90 and priced over $75 were classified as high-demand, while budget-friendly wines with scores above 95 were also highlighted. This classification scheme enabled a systematic method to organize the data, ensuring its usefulness in providing tailored recommendations and inventory management insights.

To facilitate wine demand submissions from restaurant owners, an HTML form was designed and deployed on AWS Amplify. This form allowed owners to enter specifics about their wine requirements, such as type, quantity, and preferred pricing. A Lambda function processed each submission, saving the data as CSV files in an Amazon S3 bucket for later analysis. API Gateway

was configured to enable secure data transfer from the form to Lambda, and Cross-Origin Resource Sharing (CORS) was enabled to ensure that the form's requests would interact seamlessly with the API. This setup created a convenient method for restaurant owners to communicate their inventory needs in a structured format, which was securely stored for subsequent processing.

A parallel system was also implemented for wine producers to submit wine offers that could be matched with restaurant demand. An HTML form like the restaurant form was designed to collect essential details from producers, including wine type, price, and tasting score. The form was hosted on AWS Amplify, with data processed by a Lambda function and stored as JSON files in an Amazon S3 bucket. API Gateway was configured to connect the form with the Lambda function, facilitating smooth data transmission. Through this arrangement, wine producers were able to contribute supply details that could then be evaluated against demand data from restaurants, enhancing data symmetry between supply and demand sides.

The project's data matching process was executed through a Lambda function that compared restaurant demand with producer offerings. If critical fields like wine name, type, and quantity aligned, the system logged this data to an S3 bucket, accessible to restaurant owners. If direct matches were unavailable, the system recommended high-demand wines from previously defined categories. This functionality allowed for flexible recommendations, improving inventory alignment with consumer preferences even when precise matches were not available.

Finally, Amazon SNS was integrated into the pipeline to notify restaurant owners about available matches or recommended alternatives based on demand data. These notifications were automatically triggered, allowing restaurant owners to access offer updates either through their S3 data repository or via direct email alerts. This addition streamlined communication between data points and stakeholders, ensuring that restaurant owners remained informed on inventory opportunities in real-time.

5 RESULTS

The implementation of the data pipeline resulted in a streamlined process for extracting, transforming, and storing data from Reddit and Amazon DynamoDB. By leveraging Amazon Athena for querying DynamoDB, and AWS Lambda for automating data transformations, we successfully established a real-time data pipeline. This allowed for continuous updates of data in Amazon S3, making it readily available for further analysis and visualization. Amazon QuickSight was integrated to generate visual insights, enabling efficient decision-making. The system also benefited from robust monitoring through Amazon CloudWatch, ensuring data consistency and reducing errors, while providing real-time updates for enhanced data analysis.

The development of the wine demand and supply collection system successfully met the project's goal of enabling restaurant owners and wine producers to submit and receive relevant data. The system efficiently managed data through API integrations and web scraping, collecting both restaurant and airline data. By leveraging AWS services such as Lambda, API Gateway, and S3, we facilitated real-time data processing and secure storage. The integration of SNS for notifications ensured that restaurant owners received timely updates regarding wine offers or recommended alternatives. Additionally, the system's ability to compare demand and supply data helped identify matching wine offers and automatically suggest high-demand alternatives when necessary, providing a valuable tool for both restaurants and wine producers.

Figure 3: Wine classification insight

6 CONCLUSION

The successful implementation of the data pipeline for Reddit and DynamoDB data extraction, transformation, and analysis has significantly streamlined data management and visualization processes. By utilizing AWS services such as Lambda, Athena, S3, and QuickSight, we were able to establish an automated, real-time system that ensures continuous data updates and offers valuable insights. The use of Amazon CloudWatch for monitoring ensured the stability and reliability of the system, further enhancing the overall workflow. This project demonstrates the effectiveness of leveraging cloud-based solutions to manage large datasets and generate actionable insights.

The wine demand and supply collection system proved to be a robust solution for gathering and processing data from restaurant owners and wine producers. Using AWS services, including Lambda, API Gateway, and S3, the system efficiently handled both the submission and storage of data. The integration of SNS for real-time notifications ensured that all stakeholders were promptly informed of relevant offers or recommendations. The automatic comparison between restaurant demands and wine producer offers enabled the system to provide valuable recommendations, making it an essential tool for streamlining wine sourcing for restaurants. Ultimately, this project successfully fulfilled

the objective of connecting wine producers with restaurant owners through a seamless data collection and communication platform.

Both projects highlight the power of cloud-based infrastructure in automating data collection, processing, and real-time delivery. By integrating AWS tools and APIs, we were able to develop scalable systems capable of handling dynamic and large volumes of data, providing valuable insights and recommendations. project 1 demonstrated the effectiveness of an automated data pipeline for continuous data updates, while Project 2 focused on the efficient management of wine demand and supply data. Together, these projects underscore the importance of leveraging modern cloud services to create efficient, automated, and insightful systems that benefit both businesses and their customers.

The successful implementation of both projects illustrates the transformative potential of cloud technology in streamlining operations, improving data accessibility, and enhancing decision-making. By utilizing a combination of web scraping, API integrations, and AWS services, both systems were able to address complex challenges in data collection and real-time processing. These projects not only demonstrate technical proficiency but also offer practical solutions that improve efficiency and communication between stakeholders, making them valuable contributions to their respective domains.

REFERENCES

[1] M. N. and W. J. , “Big Data: Principles and Best Practices of Scalable Real-time Data Systems,” Manning Publications Co, 2015.

[2] K. J. , “Questioning the lambda architecture,” O'Reilly Radar, 2014. [Online]. Available: https://www.oreilly.com.

[3] C. Wang, Z. Bi, and L. Da Xu, “IoT and cloud computing in automation of assembly modeling systems,” *IEEE Transactions on Industrial Informatics,* vol. 6, no. 1, p. 10, 2020.

[4] A. Gandomi and M. Haider, “Beyond the hype: Big data concepts, methods, and analytics,” *International Journal of Information Management,* vol. 35, no. 2, p. 7, 2015.

[5] A. Giachanou and F. Crestani, “Like it or not: A survey of Twitter sentiment analysis methods,” *ACM Computing Surveys (CSUR),* vol. 49, no. 2, p. 40, 2016.

[6] M. Christopher, “Logistics and Supply Chain Management,” in *Pearson,* 2016.

[7] R. T. Fielding, “Architectural Styles and the Design of Network-based Software Architectures,” University of California, Irvine, 2000.

[8] E. X. a. L. L. L. Xu, “Industry 4.0: state of the art and future trends,” *International Journal of Production Research,* vol. 56, no. 8, p. 20, 2017.

[9] D. Murray, “Tableau Your Data!: Fast and Easy Visual Analysis with Tableau Software.,” in *John Wiley & Sons,* 2016.